

COMP9319 Exercises

Solution : Please see us at the consultations if there are any questions.

A Sample XML document

```
<r>
  <a>
    <b>
      <c>
        <d>Text1</d>
      </c>
    </b>
    <b>
      <d>Text2</d>
    </b>
  </a>
  <a>
    <b>
      <c @b="1">
        <d>Text3</d>
      </c>
    </b>
    <b>
      <c>Text4</c>
    </b>
  </a>
  <a>
    <c>
      <d>Text5</d>
    </c>
  </a>
</r>
```

Question 1

What is the result when the XPath `/r/a/b/c[@b]` is evaluated on the XML document above?

Ans:

```
<c @b="1">
  <d>Text3</d>
</c>
```

Question 2

What is the result when the XPath `//c[d]` is evaluated on the XML document above?

Ans:

```
<c>
  <d>Text1</d>
</c>
<c @b="1">
  <d>Text3</d>
</c>
<c>
  <d>Text5</d>
</c>
```

Question 3

What is the result when the XPath `//a[b]//c[d]` is evaluated on the XML document above?

Ans:

```
<c>
  <d>Text1</d>
</c>
<c @b="1">
  <d>Text3</d>
</c>
```

Question 4

What is the result when the XPath `//a*/c[d]` is evaluated on the XML document above?

Ans:

```
<c>
  <d>Text1</d>
</c>
<c @b="1">
  <d>Text3</d>
</c>
```

Question 5

What is the result when the XPath `//a[*]/c[d]` is evaluated on the XML document above?

Ans:

```
<c>
  <d>Text5</d>
</c>
```

Question 6

How does XGRIND achieve data compression and yet support queries on the compressed data ?

Ans:

Compression via: mapping tags to shorter labels; compress text data via static huffman.

Queryable: content and structures are preserved.

Question 7

Calculate the min and max of the forward excess of the following tier-0 block in ISX:

((()((()((

Ans:

min forward excess: 0

max forward excess: 5